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*
*      STAAD.Pro
*      Version  2007    Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=    JAN 18, 2013
*      Time=    14:22:52
*
*      USER ID:
*****
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1. STAAD SPACE
INPUT FILE: 20121120 Pig Launcher FR.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 5 1 -5.73; 3 5 4.324 -5.08; 4 5 4.324 -5.73; 6 5 4.324 -6.38
9. MEMBER INCIDENCES
10. 1 4 3; 3 2 4; 5 6 4
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 1 5 TABLE ST H428X407X20
21. 3 TABLE ST H428X407X20
22. CONSTANTS
23. BETA 90 MEMB 3
24. **BETA 90 MEMB 3
25. MATERIAL STEEL ALL
26. SUPPORTS
27. 2 FIXED
28. ***** DEAD LOAD *****
29. LOAD 1 LOADTYPE NONE TITLE DL
30. SELFWEIGHT Y -1
31. JOINT LOAD
32. 4 FY -250
33. ***** LIVE LOAD *****
34. LOAD 2 LOADTYPE NONE TITLE FL
35. JOINT LOAD
36. 4 FX -50
37. ***** WIND LOAD X-DIRECTION*****
38. LOAD 3 LOADTYPE NONE TITLE WLX
39. MEMBER LOAD
40. 1 5 UNI GX -0.346
41. 3 UNI GX -0.346
42. ***** WIND LOAD Z-DIRECTION*****
43. LOAD 4 LOADTYPE NONE TITLE WLZ
44. MEMBER LOAD
45. 3 UNI GZ -0.364
46. ***** NOTIONAL LOAD X-DIRECTION*****
47. LOAD 5 LOADTYPE NONE TITLE NLX
48. SELFWEIGHT X -0.015
49. JOINT LOAD
50. 4 FX -3.75
51. *****NOTIONAL LOAD Z-DIRECTION*****
52. LOAD 6 LOADTYPE NONE TITLE NLZ
53. SELFWEIGHT Z -0.015
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54. JOINT LOAD
55. 4 FZ -3.75
56. ***** LOAD COMBINATION - SERVICE *****
57. LOAD COMB 7 1.0DL+1.0FL
58. 1 1.0 2 1.0
59. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
60. 1 1.0 2 1.0 3 1.0
61. LOAD COMB 9 1.0DL+1.0FL+1.0WLX
62. 1 1.0 2 1.0 4 1.0
63. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
64. 1 1.0 2 1.0 5 1.0
65. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
66. 1 1.0 2 1.0 6 1.0
67. ***** LOAD COMBINATION - ULTIMATE *****
68. LOAD COMB 12 1.0DL+1.4FL
69. 1 1.0 2 1.4
70. LOAD COMB 13 1.4DL+1.4FL
71. 1 1.4 2 1.4
72. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
73. 1 1.0 2 1.4 3 1.4
74. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
75. 1 1.0 2 1.4 4 1.4
76. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
77. 1 1.0 2 1.4 5 1.4
78. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
79. 1 1.0 2 1.4 6 1.4
80. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 4/ 3/ 1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 18

81. LOAD LIST 7 TO 11
82. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	7	-0.7689	-0.0117	0.0000	0.0000	0.0000	0.0034
	8	-0.7825	-0.0117	0.0000	0.0000	0.0000	0.0035
	9	-0.7689	-0.0111	-0.0026	0.0000	0.0000	0.0034
	10	-0.8282	-0.0117	0.0000	0.0000	0.0000	0.0037
	11	-0.7689	-0.0061	-0.0213	-0.0001	0.0000	0.0034
4	7	-0.7689	-0.0116	0.0000	0.0000	0.0000	0.0034
	8	-0.7825	-0.0116	0.0000	0.0000	0.0000	0.0035
	9	-0.7689	-0.0116	-0.0026	0.0000	0.0000	0.0034
	10	-0.8282	-0.0116	0.0000	0.0000	0.0000	0.0037
	11	-0.7689	-0.0116	-0.0213	-0.0001	0.0000	0.0034
6	7	-0.7689	-0.0117	0.0000	0.0000	0.0000	0.0034
	8	-0.7825	-0.0117	0.0000	0.0000	0.0000	0.0035

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10	-0.8282	-0.0117	0.0000	0.0000	0.0000	0.0037
11	-0.7689	-0.0174	-0.0213	-0.0001	0.0000	0.0034

***** END OF LATEST ANALYSIS RESULT *****

83. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	50.00	262.81	0.00	0.00	0.00	-166.20
	8	51.60	262.81	0.00	0.00	0.00	-169.61
	9	50.00	262.81	1.21	2.01	0.00	-166.20
	10	53.94	262.81	0.00	0.00	0.00	-179.07
	11	50.00	262.81	3.94	12.87	0.00	-166.20

***** END OF LATEST ANALYSIS RESULT *****

84. LOAD LIST 12 TO 17
85. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	12	4	0.00	1.80	0.00	0.00	0.00	0.59
		3	0.00	0.00	0.00	0.00	0.00	0.00
13	4	0.00	2.52	0.00	0.00	0.00	0.00	0.82
		3	0.00	0.00	0.00	0.00	0.00	0.00
14	4	0.00	1.80	-0.31	0.00	0.00	0.10	0.59
		3	0.00	0.00	0.00	0.00	0.00	0.00
15	4	0.00	1.80	0.00	0.00	0.00	0.00	0.59
		3	0.00	0.00	0.00	0.00	0.00	0.00
16	4	0.00	1.80	-0.04	0.00	0.00	0.01	0.59
		3	0.00	0.00	0.00	0.00	0.00	0.00
17	4	0.04	1.80	0.00	0.00	0.00	0.00	0.59
		3	0.00	0.00	0.00	0.00	0.00	0.00
3	12	2	262.81	0.00	70.00	0.00	-232.68	0.00
		4	-253.60	0.00	-70.00	0.00	0.00	0.00
13	2	367.94	0.00	70.00	0.00	-232.68	0.00	0.00
		4	-355.04	0.00	-70.00	0.00	0.00	0.00
14	2	262.81	0.00	72.24	0.00	-237.45	0.00	0.00
		4	-253.60	0.00	-70.63	0.00	0.00	0.00
15	2	262.81	1.69	70.00	0.00	-232.68	2.82	0.00
		4	-253.60	0.00	-70.00	0.00	0.00	0.00
16	2	262.81	0.00	75.52	0.00	-250.70	0.00	0.00
		4	-253.60	0.00	-75.33	0.00	0.00	0.00
17	2	262.81	5.52	70.00	0.00	-232.68	18.02	0.00
		4	-253.60	-5.33	-70.00	0.00	0.00	0.00
5	12	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	1.80	0.00	0.00	0.00	-0.59
13	6	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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14	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	1.80	-0.31	0.00	-0.10	-0.59
15	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	1.80	0.00	0.00	0.00	-0.59
16	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	1.80	-0.04	0.00	-0.01	-0.59
17	6	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.04	1.80	0.00	0.00	0.00	-0.59

***** END OF LATEST ANALYSIS RESULT *****

86. LOAD LIST ALL
87. PARAMETER 1
88. CODE BS5950
89. BEAM 2 ALL
90. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.002	13
		0.00	0.00	0.82	0.00
3 ST	H428X407X20	PASS	ANNEX I.1	0.330	16
		262.81 C	250.70	0.00	-
5 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.002	13
		0.00	0.00	-0.82	0.65

***** END OF TABULATED RESULT OF DESIGN *****

91. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 18,2013 TIME= 14:22:53 ****

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